Building Denmark's power system from scratch

Investment

These solutions are written to be read after the workshop, one subsection per question. The keys of the closed questions are given in the worksheet's lettering. Each open question has a model answer, which is one good answer among several, as in the Model Lab. Figures are ESM's own from section 7, or drawn from the Explorer's data, and each caption says which. Explorer views are named as on the worksheet, as tab › figure, and linked. Where the evidence is a number the figure does not draw, the pointer says so: press *Table* at the figure's top right, which lists every number behind it. *Headline numbers* shows eight quantities as tiles; the others are in its table.

1 Read the reference system

Part 1 · ESM §7.2–7.3

1. Optimal investment

(i) Result 6. Differentiating generator g's terms of the Lagrangian (ESM eq. (41)) with respect to its output in one hour gives, with z the zone of g,

$$q_{g,h}: \quad \lambda_{z,h} - c_g - \tau e_g + \underline{\mu}_{g,h} - \overline{\mu}_{g,h} = 0,$$

with complementary slackness $\underline{\mu}_{g,h}\, q_{g,h} = 0$ and $\overline{\mu}_{g,h}\,(\gamma_{g,h}\bar{q}_g - q_{g,h}) = 0$. Take the three cases in turn. If the price is above the running cost, carbon included, then $\overline{\mu}_{g,h} > \underline{\mu}_{g,h} \geq 0$: the plant runs at its limit, so it is not at zero, $\underline{\mu}_{g,h} = 0$ and $\overline{\mu}_{g,h} = \lambda_{z,h} - c_g - \tau e_g$. If the price is below it, $\underline{\mu}_{g,h} > 0$: the plant is off, its limit is slack and $\overline{\mu}_{g,h} = 0$. At equality both are zero. In every hour, therefore,

$$\overline{\mu}_{g,h} = \max\left(\lambda_{z,h} - c_g - \tau e_g,\ 0\right).$$

The capacity $\bar{q}_g$ enters the Lagrangian through its annual fixed cost F_g and through every hour's limit $\gamma_{g,h}\bar{q}_g$, so its condition (ESM eq. (42)) is

$$\bar{q}_g: \quad -F_g + \sum_h \gamma_{g,h}\,\overline{\mu}_{g,h} + \underline{\beta}_g - \overline{\beta}_g = 0.$$

Substituting the first result and rearranging gives Result 6. The sum is the scarcity rent a MW of g earns in a year: in each hour, its availability times the margin of the price over its running cost, carbon included, counted only in the hours it is worth running. The

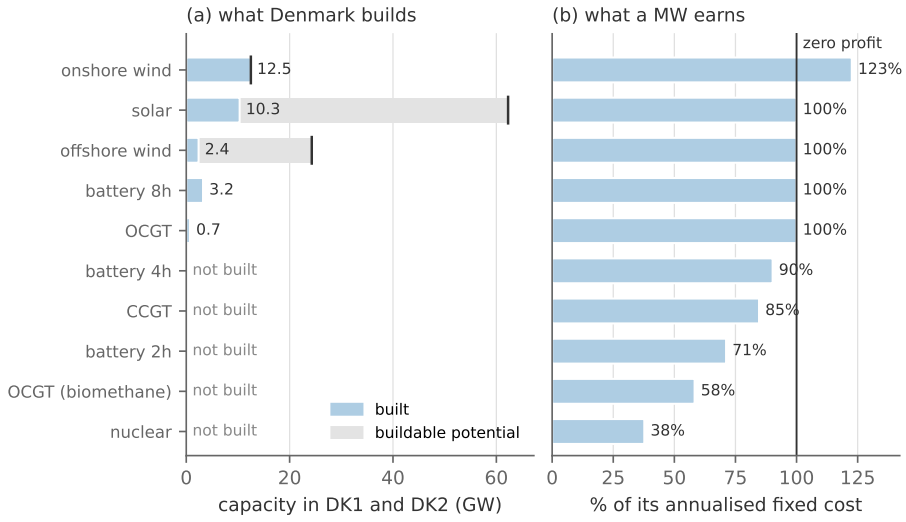


Figure S1: ESM, Fig. 7.2: (a) what Denmark builds in the reference case, DK1 and DK2 together, against the potential; (b) what a MW of each candidate earns as a share of its annual fixed cost.

left-hand side is then what a MW earns beyond its fixed cost: a land rent if positive, a shortfall if negative.

(ii) Key: (a) and (b). $\overline{\beta}_g$ is the multiplier on the potential, positive only where the potential binds: onshore wind, in both zones. A MW there earns more than its fixed cost, and the excess is the land rent (onshore wind earns 118 % of its fixed cost in DK1 and 145 % in DK2). Solar and offshore wind, in both zones, and the DK2 peaker are built below their potentials (the peaker has none), so both multipliers are zero and a MW earns exactly its fixed cost. Nuclear is not built: there $\underline{\beta}_g > 0$, since a MW would earn little more than a third of its fixed cost (38 %, Figure S1). Which of the three cases a technology is in is the first thing to establish about any capacity result. In the Explorer: the potentials are the ticks in Overview › *What Denmark builds*, zone by zone with the zone menu; what a MW earns against its fixed cost is Money › *Rent per MW against annual fixed cost*, where the bars of the technologies built below their potential end at 1, onshore wind's pass it, and nuclear's faded bar stops well short.

(iii) Capture price and levelised cost. Result 6 can be read per MWh. Let x_g be the MWh a MW of g produces in a year. Its rents are its output times the margin of its capture price over its running cost, $x_g(CP_g - c_g - \tau e_g)$, and by the definition of the levelised cost (ESM eq. (39)) its fixed cost is $x_g(LCOE_g - c_g)$, both at the hours it runs. Dividing Result 6 by x_g,

$$CP_g - \left(LCOE_g + \tau e_g\right) = \frac{\overline{\beta}_g - \underline{\beta}_g}{x_g},$$

so the three cases of Result 6 are three relations between the capture price and the levelised

Table 1: ESM, Table 5: levelised cost and capture price in the reference system. The levelised cost excludes the carbon price; the gas turbine's carbon cost of 39 €/MWh has to be added to compare it with its capture price.

Technology	Capacity GW	Full-load hours h/yr	LCOE €/MWh	Capture price €/MWh	Value factor	
solar	10.3	953	30.8	30.8	0.58	interior
onshore wind	12.5	3,060	32.0	39.0	0.73	at its cap
offshore wind	2.4	4,508	37.3	37.3	0.70	interior
OCGT	0.7	938	95.0	134.2	2.51	interior

cost, carbon included (Table 1). Built below its potential, a technology captures exactly its levelised cost: offshore wind and solar, in both zones, and the gas peaker once its carbon cost is counted. At its potential it captures more, by the land rent per MWh: onshore wind, the only technology at its potential (39.0 against 32.0 €/MWh). Not built, it would capture less than its levelised cost: nuclear, the combined cycle, the shorter batteries. The levelised cost here is at the hours the technology runs in the solution, an outcome of the model rather than an input. In the Explorer, the two prices of each technology built are in Money › *Levelised cost and capture price*; what a MW of a technology not built would earn, in the faded bars of Money › *Rent per MW against annual fixed cost.*

2. The generation mix

(i) Key: (c). At its potential a technology captures its levelised cost plus the land rent per MWh (question 1(iii)); below it, it captures its levelised cost and no more. Onshore wind costs more per MWh than solar and still keeps part of each MWh as rent, where solar keeps none, so its output must sell for more (39.0 against 30.8 €/MWh). Both sell at the same Danish prices, so onshore wind's value factor is the higher one (0.73 against 0.58; Money › *Levelised cost and capture price* gives each value factor under the technology's name): the wind blows in winter, at night and in the evening, when demand is high and solar is absent, while solar produces when every panel in Denmark and Germany does. The hours a technology runs are already in its levelised cost, and the land rent is what the scarce sites are worth, a result of the prices rather than a cost.

(ii) Key: (b). Result 6 with both multipliers zero and $\gamma_h = 1$ says that the peaker's rents equal its fixed cost,

$$\sum_h \max(\lambda_h - c,\, 0) = F.$$

Only the hours with a price above the running cost count: below it the peaker stays off and loses nothing. The condition is about how far and how often the price rises above c, not about its average.

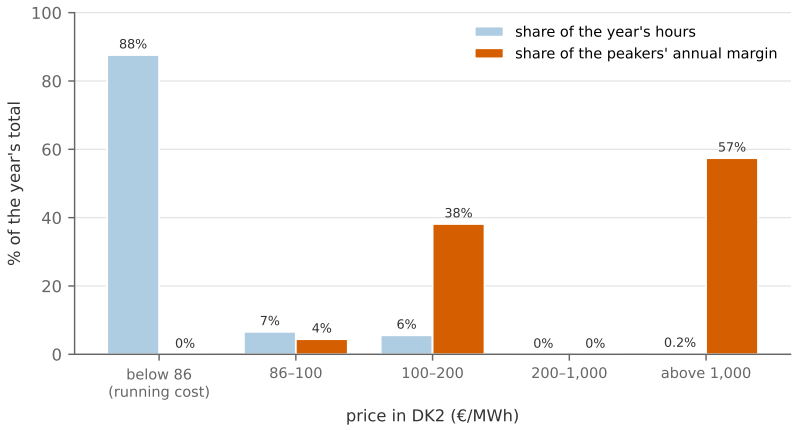


Figure S2: ESM, Fig. 7.3: the hours of the year in bands of the DK2 price, and the share of the turbines' annual operating margin earned in each.

(iii) Gas turbines. A gas turbine is paid for being available when power is scarce, not for the energy it makes. It runs in winter spells with weak sun and high demand, when imports are limited and the batteries run empty (about 938 full-load hours, all between October and March). A MW earns the margin of the price over its running cost in those hours, and most of it in a few: the 21 hours above 1,000 €/MWh bring more than half of it (57 %, Figure S2). Turbines are added until one more MW would no longer earn its fixed cost: in DK2 a MW earns exactly that, while in DK1 it would earn just under it and none is built. Being cheap to build, the turbine is the cheapest way to stand by for long, rare shortfalls, which a battery could only cover with enormous energy capacity.

In the Explorer, reference case: when the turbine runs, in Operation › *A week of operation*, zone DK2, stepping through the weeks; its full-load hours, under its name in Money › *Levelised cost and capture price*; its running cost, in Money › *Running costs, and demand's willingness to pay*; which hours pay for it, in Money › *Peakers' margin by price band*. The hours above 1,000 €/MWh are too few to show as a bar there: count them on Overview › *The price-duration curve*, zone DK2, log scale, or press *Table* on the price-band figure. That a MW earns its fixed cost in DK2 and just under it in DK1 is in Money › *Rent per MW against annual fixed cost*.

3. What sets the price

(i) Key: (b) and (d). Trade sets the Danish price in about seven hours in ten, in both zones, and wind and solar in surplus in almost all of the rest, in every month and most often in May and June; even then, trade is at the margin more often than surplus. A Danish gas plant is at the margin in about 110 hours in DK2, around 1 % of the year, all of them between October and March, and even then in a small share of the winter's hours (Figure S3; in the Explorer the figure's legend gives each category's hours a year, and the

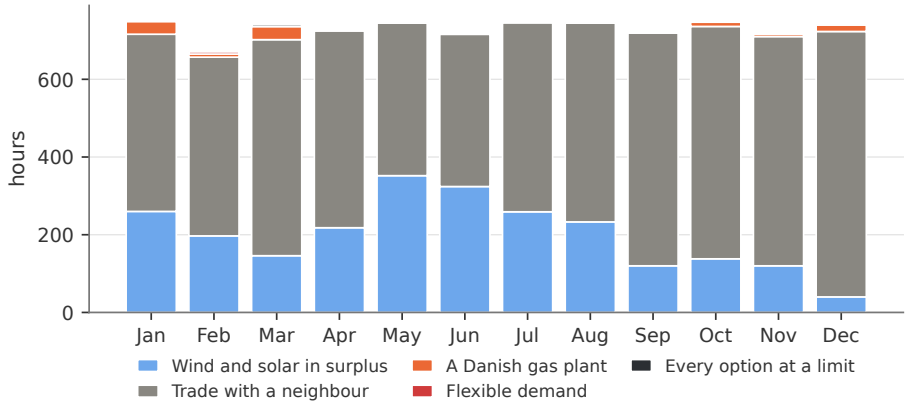


Figure S3: Drawn from the Explorer's data, as its Operation › *What is at the margin, month by month* draws it: the hours of each month in DK2 in the reference case, by what is at the margin of the price.

zone menu switches to DK1). Denmark builds a system of its own, but most of the time its price is made abroad.

(ii) Key: (d). An interconnector that carries power below its capacity ties the prices at its two ends, losses apart, so a Danish zone shares its price with the market behind it, and a plant abroad sets it. Only when the lines are full does Denmark price on its own: at zero when it cannot export its surplus wind and solar, and at its own peaker or its flexible demand when it cannot import enough, which in DK2 happens in about 115 hours of the year. The price is a marginal cost, and the margin is mostly abroad; the Danish peaker still runs, about 938 full-load hours a year (under its name in Money › *Levelised cost and capture price*), mostly at prices set elsewhere.

(iii) Scarcity. In the hours with the highest prices, in mid-January, every Danish option is at a limit: the gas peaker runs flat out, flexible demand is fully curtailed, the batteries spread what they hold over the spell, and the lines from DK1 and Germany carry all they can. DK2 passes power on to southern Sweden, which is shorter still, at almost the same price (Operation › *One hour, inspected*, zone DK2, which opens on the segment with the year's highest price: the peaker's bar is full and the batteries discharge; the other zones' prices in that segment are in its table, so press *Table*). The note under Operation › *What is at the margin, month by month* explains why the Explorer counts such an hour as trade: no Danish option is part-loaded, and a line is below its capacity. No running cost is at the margin, so the price rises until it rations what is left. In the long-run optimum its level is pinned down by the investment condition: high enough, in few enough hours, that the peakers and batteries standing by for them just pay for themselves, in Denmark and in southern Sweden, whose price DK2 shares in these hours. The 21 hours above 1,000 €/MWh bring the Danish peakers more than half of their margin (Figure S2). Were

more capacity built, such hours would be fewer and their prices lower.

Stretch question

(i) The battery's conditions. The discharge condition gives $\overline{\phi}_h^d - \underline{\phi}_h^d = \lambda_h - \theta_h/\eta_d - c^s$. If the price is above $\theta_h/\eta_d + c^s$, the right-hand side is positive, so $\overline{\phi}_h^d > 0$ and the battery discharges at full power; then $\underline{\phi}_h^d = 0$ and $\overline{\phi}_h^d = \lambda_h - \theta_h/\eta_d - c^s$. Otherwise $\overline{\phi}_h^d = 0$. So

$$\overline{\phi}_h^d = \max\left(\lambda_h - \theta_h/\eta_d - c^s,\ 0\right),$$

the price less what the energy drawn from the battery is worth, in the hours it discharges flat out. The same steps on the charge condition give

$$\overline{\phi}_h^c = \max\left(\eta_c\,\theta_h - c^s - \lambda_h,\ 0\right),$$

what the energy put in store is worth less the price paid, in the hours it charges flat out. The state-of-charge condition gives $\overline{\psi}_h - \underline{\psi}_h = \theta_{h+1} - \theta_h$, and the battery cannot be full and empty at once, so

$$\overline{\psi}_h = \max\left(\theta_{h+1} - \theta_h,\ 0\right),$$

positive only when the battery is full: the rise in what a MWh in store is worth from one hour to the next. The value of stored energy is itself set by prices: in an hour the battery charges below its rating, $\theta_h = (\lambda_h + c^s)/\eta_c$, and in one it discharges below its rating, $\theta_h = \eta_d(\lambda_h - c^s)$ (ESM's Result 3, the arbitrage band). All three multipliers are therefore differences between the prices of different hours, net of losses and operating costs: the price swings the battery trades on.

(ii) What another MW earns. For a battery that is built, $\underline{\beta}_s = 0$, so another MW would earn $\sum_h(\overline{\phi}_h^c + \overline{\phi}_h^d) + \bar{\delta}\sum_h \overline{\psi}_h$ a year, exactly its fixed cost F_s. It earns on its power in the hours it charges or discharges flat out, when more power would let it move more energy from a cheap hour to an expensive one, and on its energy, $\bar{\delta}$ times over, in the hours it is full, when more storage would let it carry more cheap energy into expensive hours. Both are price differences between hours: what a MW earns is set by how wide the price swings are and how often they come, not by any running cost. In the reference these are the daily swings of spring and summer, cheap at midday and expensive in the evening and night, and a few winter hours with very high prices (Operation › *The price through the day, month by month*, and the left end of Overview › *The price-duration curve*).

(iii) Why 8 hours. At the reference prices a 2-hour battery would earn about seven tenths of its annual fixed cost and a 4-hour one about nine tenths (71 % and 90 %), while the 8-hour one earns all of it (Money › *Rent per MW against annual fixed cost*).

A MW earns roughly in proportion to its hours of storage, because the swings it trades on are long: from spring to late summer the price is low through the middle of the day and high through the evening and night, so the batteries charge for many hours and discharge for many (Operation › *What the batteries do through the day, month by month*, and *The price through the day, month by month*). The cost does not rise in proportion: the inverter costs the same per MW whatever the duration, and only the cells scale with it (about 9.7 k€ per MW and 6.4 k€ per MWh a year, from the three batteries' fixed costs, which no figure draws: press *Table* on Money › *Rent per MW against annual fixed cost*). An hour of storage therefore costs least in the longest battery on the menu.

2 Shock it: carbon, fuel and capital

Part 2 · ESM §7.4

1. A higher carbon price

Prediction. Danish emissions: *falls*; solar built in Denmark: *rises*; batteries: *rises*; gas peakers: *falls*; the average Danish price: *rises*.

Danish emissions fall to zero, and the gas peaker disappears: at 225 €/t the same turbine is cheaper to run on biomethane, which is built instead. Solar and batteries grow, because a higher carbon price raises the price in exactly the hours when fossil plants used to set it, and those are the hours solar and storage can serve. The average price rises by about a tenth, far less than the carbon price: in a system this renewable, carbon reaches the price only in the few hours when an emitter is at the margin. Figure S4 shows the mix along the whole sweep; the Explorer compares the two scenarios directly (Overview › *What Denmark builds*, *Carbon price 225 EUR/t* against the reference; the price is the tile "Danish price" in Overview › *Headline numbers*).

2. Denmark's marginal abatement cost curve

The shape. The system abates every tonne that costs less than the carbon price, so the price at which a tonne goes is its marginal abatement cost: the curve is the sweep with its axes swapped, abatement being the fall in emissions (Figure S5). It starts low and flat: most Danish emissions go below 75 €/t, spread fairly evenly over that range, as solar, wind and batteries take over the gas turbines' energy and leave them as backup. It then turns steep: what remains is backup in winter spells, when solar does little and the batteries run empty, so each tonne costs more than the last, and emissions fall only slowly as the price rises further. Near 220 €/t comes a flat step: the turbines switch to

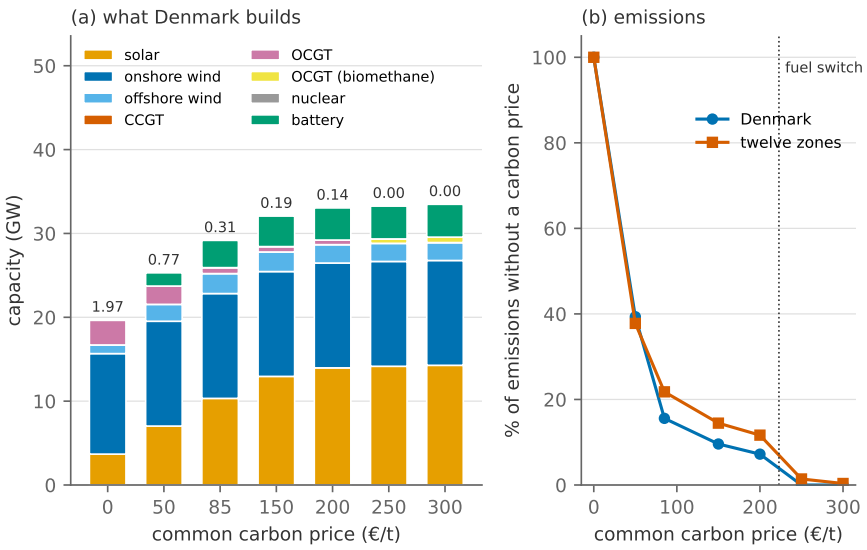


Figure S4: ESM, Fig. 7.5: the common carbon price swept from zero to 300 €/t: (a) what Denmark builds, with Danish emissions in Mt above each bar; (b) Danish and twelve-zone emissions, each as a share of its level without a carbon price.

biomethane, and the remaining tonnes go at once. Then nothing is left, and the curve ends.

The height of the step. The same turbine run on biomethane instead of gas costs $c_b - c_g$ more per MWh and avoids e tonnes, so every remaining tonne costs the same to abate,

$$\frac{c_b - c_g}{e} = \frac{149.4 - 46.8}{0.46} \approx 223 \text{ €/t}.$$

Beyond the question. The curve is the Danish part of a twelve-zone optimum, not what abatement would cost Denmark alone: every zone pays the carbon price, so the Danish part need not fall at every step: Danish emissions are even slightly higher at the reference 85 €/t than at 75. Above the switch the twelve zones' emissions also fall almost to nothing, since the neighbours' gas plants may burn biomethane too (Figure S4(b)).

3. Expensive gas

(i) Prediction. Gas peakers built in Denmark: *rises*; combined-cycle plants built in Germany: *falls*; Danish emissions: *falls*; flexible demand left unserved in Denmark: *rises*.

Denmark builds about twice the peaking capacity of the reference, yet burns less gas: its peakers run about a third as many full-load hours, and Danish emissions fall by about a quarter, and those of the twelve zones slightly. Germany builds no new combined-cycle plants at all, and flexible demand in Denmark is curtailed about half as much again. More gas capacity and less gas burnt: (ii) and (iii) explain how the optimum does both

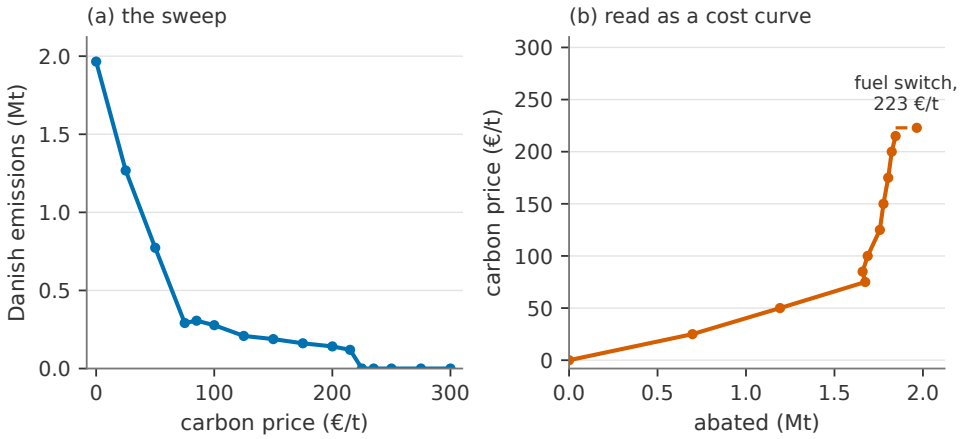


Figure S5: Drawn from the Explorer's data: (a) its sweep of Danish emissions against the common carbon price, each point a solved scenario; (b) the same points with the axes swapped, abatement counted from the emissions without a carbon price. The dashed step is the fuel switch at $(c_b - c_g)/e$.

(Figure S6). In the Explorer, *Gas at 2 x, new plants only* against the reference: the peakers in Overview › *What Denmark builds*, and Germany's combined cycles in the same figure with the zone set to Germany; their full-load hours under the technology's name in Money › *Levelised cost and capture price*; the flexible demand not served is not among the tiles of Overview › *Headline numbers*, so press *Table* there for "Flexible demand not served, Denmark".

(ii) Key: (a). A screening curve compares plants for a given duty, the hours a year a MW has to run, and the shock changes the duty. A higher fuel price adds to a plant's annual cost in proportion to the hours it runs: at the hours Germany's combined-cycle plants run in the reference, doubling gas adds more to a MW's annual cost than its whole fixed cost. For energy over that many hours there are substitutes whose cost does not move with gas: wind, solar, and batteries to carry solar into the night. Once they take the long hours, new gas plants are left with standing by for the winter spells, a duty of a few hundred hours (Denmark's peakers now run about 350 full-load hours, against about 938), and at that duty the peaker is the cheaper plant. The crossover moved towards the combined cycle; the duty moved further, towards the peaker (ESM Fig. 7.1; in the Explorer, Money › *Screening curves, and where each technology landed* shows both plants' curves and where they cross in each scenario, and the peakers' full-load hours are under their name in Money › *Levelised cost and capture price*).

(iii) Key: (c). In the reference, Germany's combined-cycle plants run at full available output in almost every hour the Danish peaker runs (Operation › *One hour, inspected*, zone Germany, which opens on the segment with the year's highest price: the combined-

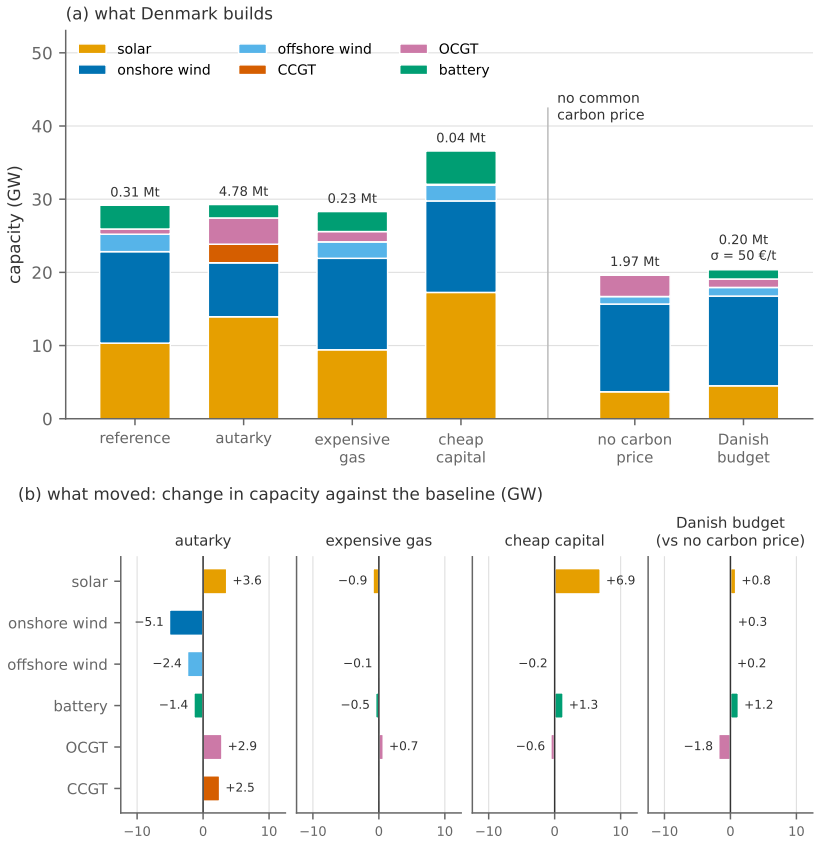


Figure S6: ESM, Fig. 7.6: (a) what Denmark builds under autarky, expensive gas and cheap capital, and under a Danish budget beside the system with no carbon price; (b) the change in capacity against each experiment's baseline.

cycle bar is filled to what could have run). They were built for the energy they produce over thousands of hours, and their standing by in winter came free with it. Expensive gas takes away the energy job, and the standing by goes with it. A peaker earns its keep only in hours when the price rises above its running cost because capacity is short. A higher gas price pays it nothing extra in the hours it sets the price itself, since its own cost rises as much; but with the combined-cycle plants gone, scarce hours come more often, and the model adds peakers until a MW again earns just its fixed cost: Denmark's peaking capacity about doubles, to about 1.4 GW. Scarcity becomes more frequent and less extreme: DK2's price passes 200 €/MWh about three times as often, while the year's highest price falls by more than half, because twice the peaking capacity stands by (Overview › *The price-duration curve*, zone DK2, log scale).

(iv) Key: (a) and (c). Expensive gas lifts the peaker's running cost above flexible demand's willingness to pay (about 128 against 100 €/MWh; Money › *Running costs, and demand's willingness to pay*), so the first margin is on the demand side: in about half of the hours flexible demand now steps back, no peaker runs at all (the DK2 curve in Overview › *The price-duration curve* spends hundreds of hours between the lines "Flexible

demand's valuation" and "Gas peaker's running cost"). What the combined-cycle plants produced comes mostly from German solar, now built up to its potential, with more German batteries to move it into the evening and night; coal and lignite make up less than a fifth of it (Overview › *Energy balance* and *What is built*, zone Germany, against the reference; the tick on solar's bar is its potential). In Denmark, solar, batteries and offshore wind all come out somewhat lower than in the reference (Overview › *What Denmark builds*), because Germany's new solar and batteries now trade on the same price swings. Which margin replaces gas depends on which of its jobs is meant: its energy goes to solar and storage, its last few hours to demand response, and its standing by in winter stays with gas, in more Danish peakers.

4. Cheap capital

(i) Prediction. Solar built in Denmark: *rises*; batteries: *rises*; gas peakers: *falls*; the average Danish price: *falls*; the Danish net cost: *falls*.

Solar, wind and batteries carry almost their whole cost as an annuity, so a lower discount rate makes them much cheaper, while a peaker's cost is largely fuel: more solar and storage, and only a few peakers (Figure S6). The average price falls by about three tenths. In a system whose prices recover capital, the cost of capital *is* the price level: the whole price-duration curve sits lower (Overview › *The price-duration curve*, *Discount rate 2 %* against the reference), because every technology that sets the price in long-run equilibrium now needs less rent to break even. Almost the whole Danish bill is capital (95 % of it in the reference), which is why the price falls about as far as the bill does (31 %). The price is the tile "Danish price" in Overview › *Headline numbers*; no figure draws the capital share, so press *Table* there for "Share of the Danish bill that is capital".

(ii) The discount rate. The annual cost of a wind turbine or a solar park is almost all annuity, because its fuel is free, and the annuity rises with r (ESM eq. (37)). A gas plant's annual cost is largely fuel, which is paid by the hour and does not depend on r. A higher discount rate therefore raises the cost of the capital-intensive technologies the transition builds relative to the fuel-intensive ones it replaces. In the Explorer, the table of Overview › *What Denmark's system costs* (press *Table*) splits the Danish bill into capital and operating cost by technology: almost all of it is capital. The bars themselves do not show the split.

3 Margins of response

Part 3 · ESM §7.4, and beyond

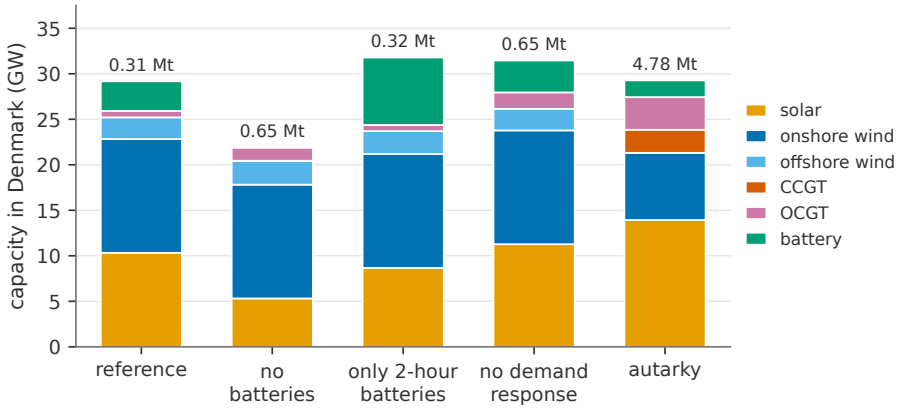


Figure S7: Drawn from the Explorer's data: what Denmark builds in the reference case and in the four scenarios of this part that take a margin of response away, with Danish emissions above each bar. ESM's technology colours; batteries are counted in GW of power.

1. No batteries

(i) Prediction. Gas peakers built in Denmark: *rises*; flexible demand not served: *falls*; solar built in Denmark: *falls*; Danish emissions: *rises*.

Without storage Denmark builds about twice the peaking capacity and only about half the solar: there is nothing to move solar's noon surplus into the evening (Figure S7). The surprise is demand response, which is called on less, not more, the subject of (ii): about a third less flexible demand goes unserved, and the DK2 price reaches the flexible block's valuation in about 300 hours instead of about 500. Danish emissions roughly double, and the twelve-zone system costs a few per cent more: removing an option can never make the optimum cheaper.

In the Explorer, *No batteries can be built* against the reference: what is built, in Overview › *What Denmark builds*; the hours at the flexible block's valuation, where the DK2 curve in Overview › *The price-duration curve* crosses the line "Flexible demand's valuation". Danish emissions are a tile of Overview › *Headline numbers*; the flexible demand not served, the hours at its valuation and the twelve-zone system cost are not, so press *Table* there for "Flexible demand not served, Denmark", "Hours DK2's price reaches the flexible block's valuation" and "Cost of the whole 12-zone system".

Operation › *What is at the margin, month by month*, zone DK2, shows flexible demand at the margin of the price far *more* often without batteries (about 75 hours a year against 6, in the legend; in DK1 it falls, from about 35 to about 15). Being at the margin and being curtailed are different things. Flexible demand is at the margin only when part of the block is curtailed, so that the price sits at exactly 100 €/MWh. In the reference nearly every hour that curtails flexible demand curtails all of it, and the price goes above

100 €/MWh, set together with a neighbour; without batteries fewer hours get that far, and more of those that do stop at 100 €/MWh.

(ii) The mechanism. Without storage, the tight hours the batteries used to cover must be covered by firm capacity, so Denmark builds more gas peakers, about twice as many. A peaker's running cost (about 86 €/MWh with carbon) is below the flexible block's willingness to pay (100 €/MWh). With more peakers standing by, many hours that used to clear at the flexible block's willingness to pay now clear at the peaker's running cost: running the peaker is cheaper than leaving flexible demand unserved, so less of it goes unserved, in fewer hours. Margins of response are substitutes, and removing one changes which of the others is at the margin. In the Explorer: the two costs in Money › *Running costs, and demand's willingness to pay*; the longer step at "Gas peaker's running cost" and the shorter stretch above "Flexible demand's valuation" in Overview › *The price-duration curve*, zone DK2; and a Danish gas plant at the margin of the DK2 price in about six times as many hours, in the legend of Operation › *What is at the margin, month by month.*

2. Batteries and peakers

(i) Prediction. Battery power built in Denmark: *rises*; gas peakers: *about the same*; solar: *falls*; emissions in all twelve zones: *rises.*

With only short batteries on offer, Denmark buys power and gives up energy: more than twice the battery power, but only a little over half the energy in store. The peakers barely move (slightly fewer): the extra battery power covers the evening peaks, and the long winter spells, which neither battery can carry through, still fall to the peakers. Less energy in store moves less of solar's midday output into the evening and night, so less solar is built, in Denmark and still more in Germany, and gas burns in its place: emissions across the twelve zones rise by about a quarter. The 8-hour battery was the model's choice, not a default, and without it the system costs more. In the Explorer, *Only 2-hour batteries can be built* against the reference: battery power and solar in Overview › *What Denmark builds*, or the tile "Batteries" in Overview › *Headline numbers*; the energy in store and the twelve-zone emissions are not among the tiles, so press *Table* there for "Batteries built in Denmark (energy)" and "Emissions, all 12 zones"; Germany's solar, in Trade and emissions › *What the neighbours add.*

(ii) Key: (b) and (c). In summer the batteries move solar's midday output into the evening and night, every day, and no peaker is needed. In the winter week the peaker runs without stopping for more than a day, twice, while the batteries discharge only three times, once into the year's highest price: a shortfall that lasts more than a day needs more energy than an 8-hour battery holds. The rest of the year is the same: the batteries charge and discharge every day from spring to late summer and much less in winter (Operation

› *What the batteries do through the day, month by month*, zone DK2), and the peaker runs only from October to March (Operation › *A week of operation*, zone DK2, stepping through the weeks with the week menu).

(iii) Key: (d). A battery's cost is almost all in building it, and its energy capacity must hold the whole shortfall; running it costs almost nothing. It pays where a short shortfall comes back day after day, as in A, and the same energy capacity is used again and again. A peaker is cheap to build but burns fuel for every MWh it makes, so it pays where shortfalls are long and rare, as in B, for which a battery would need storage several days long, used once a year. (Covering 1 MW costs a battery about 35 k€ a year in A, against about 170 k€ for a peaker; in B about 470 k€, against about 50 k€. The peaker's fixed and running cost are where its line starts and how steeply it rises in Money › *Screening curves, and where each technology landed*; the batteries' fixed costs are drawn in no figure, so press *Table* on Money › *Rent per MW against annual fixed cost*.) The reference has both kinds of shortfall, so it builds both.

3. Demand response

(i) Prediction. The highest Danish price of the year: *rises*; gas peakers: *rises*; batteries: *rises*.

Denmark builds about two and a half times the peaking capacity and somewhat more storage: the hours that demand response used to clear must now be cleared by machines held in reserve for them (Figure S7). More of the tight hours clear at the peaker's running cost and fewer at or just above 100 €/MWh, while the scarcity hours come more often and the highest price of the year rises by about a sixth. No hour reaches the value of lost load even now. In the Explorer, *No demand response* against the reference: the highest price is the left end of Overview › *The price-duration curve*, zone DK2, log scale; peakers and batteries are tiles of Overview › *Headline numbers*; that no hour reaches the value of lost load is not, so press *Table* there for "Hours at the value of lost load, worst Danish zone".

(ii) Key: (b). In the reference, Danish flexible demand is itself at the margin of the DK2 price in very few hours: trade sets the price in most hours, and wind and solar in surplus in most of the rest (Figure S3; in the Explorer its legend gives the hours a year). Yet the price distribution moves when demand response goes, because Denmark builds about two and a half times the peaking capacity: more tight hours clear at the peaker's running cost, the hours above 130 €/MWh double, and the year's highest price rises (Overview › *The price-duration curve*, DK2 on a log scale, *No demand response* against the reference: the step at "Gas peaker's running cost" is longer). The point of Brown et al. (2025) survives in this form: a little demand elasticity changes the firm capacity a system builds, and with it the shape of its prices, even where the price is mostly made abroad.

4. Denmark's borders

(i) Prediction. Offshore wind built in Denmark: *falls*; gas peakers: *rises*; Danish emissions: *rises*; the Danish net cost: *rises*.

Interconnection is a margin of response in its own right (Chyong et al. 2024). On its own, Denmark must cover its own calm, dark hours, so it builds gas, combined cycles as well as peakers, and it has nowhere to sell surplus wind: offshore wind is no longer worth building at all, and onshore wind, at its potential in the reference, falls by about two fifths, all of it in DK1 (Figure S7). Danish emissions rise some fifteenfold, to several million tonnes, and the Danish bill by about an eighth: openness is worth something, and the difference between the two bills, net of trade, is its price (the Danish system costs 12 % more on its own, ESM §7.4). In the Explorer, *Autarky* against the reference: what is built in Overview › *What Denmark builds*; emissions and the bill are the tiles "Danish emissions" and "Danish net cost" of Overview › *Headline numbers*.

(ii) Prediction. Danish emissions: *falls*; emissions of all twelve zones: *about the same*; Danish net exports: *falls*; solar built in Denmark: *rises*.

Danish emissions fall as required, while the twelve-zone total barely moves. A part of the Danish cut (about three tenths) is achieved by exporting less and importing more power made elsewhere, with fossil fuels: under a territorial budget, imports count as carbon-free. The budget's shadow price is far below the common carbon price that would bring Danish emissions as low (50 against 150 €/t), let alone the one the fuel switch needs (223 €/t): shifting production across the border is the cheapest way to meet a territorial target (Figure S6). In the Explorer, *Danish budget at 10 %* against *Carbon price 0 EUR/t*: zone by zone in Trade and emissions › *Where the emissions are*; the tiles "Danish emissions", "Emissions abroad" and "Net imports" of Overview › *Headline numbers*; the shadow price is not among the tiles, so press *Table* there for "Shadow price of the Danish budget".

(iii) Key: (c). The cheapest tonne to remove from Danish territory is the one moved abroad, and a loose budget needs only those: it is met by importing more and running Danish peakers less. A tight budget exhausts them and has to build: solar, wind and batteries enter, and they run in hours when they displace fossil generation in the neighbouring zones too (the share of the cut that leaks falls from nearly all of it at half the unpriced emissions to about three tenths at a tenth, 31 %, and almost none at a twentieth). Leakage is not a fixed property of a policy; it depends on which margin meets it. In the Explorer, compare *Danish budget at 50 %*, *25 %*, *10 %* and *5 %* in turn with *Carbon price 0 EUR/t* in Overview › *Headline numbers*: the change in the tile "Danish emissions" against the change in "Emissions abroad", and the tiles "Net imports" and "Batteries";

the solar and wind built are in Overview › *What Denmark builds.*

Beyond the question: margins as substitutes. Read across the part, Figure S7 makes one point four times. Take a margin of response away, and the others carry the adjustment: peakers for batteries and for demand response, power for energy when the battery is short, gas and solar for trade. Each substitute costs more than what it replaces, so the system's cost rises every time, and what carries the adjustment decides what happens to emissions.